

# Radiotherapy of Choroidal Melanoma with Iodine-125

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**Abstract:** The low-energy radioactive source iodine-125 is ideal for the irradiation of an intraocular tumor such as a choroidal melanoma. The energy (28 kev) of iodine-125 should be sufficient to allow sterilization of tumors as high as 10 mm. However, shielding of 28-kev photons can be achieved with thin metal (eg, gold) plaques. These eye plaques can be individually designed for each eye tumor, depending on its size, shape, and location. This paper deals with the basic considerations and preliminary observations that have lead us to the adoption of iodine-125 as the sole source for the radiation treatment of all intraocular tumors. [Key words. brachytherapy, choroidal melanoma, iodine-125, radiotherapy] *Ophthalmology* 87:582-590, 1980

In order to appreciate the advantages of iodine-125 as the radiation source for the treatment of intraocular tumors, it is necessary to discuss briefly the history of radiotherapy of ocular tumors with other types of radioactive materials. These include radon, cobalt, and more recently, ruthenium.

Radon seeds were used first by Moore,<sup>1,2</sup> but the radiation total dose and dose rate are unclear. Several authors report experiences with radon.<sup>1-7</sup> The total dose ranged from 3,500 rads

to 13,500 rads; no dose rates were given. Radon is a powerful isotope having an energy range of 0.22-2.20 mev, an average photon energy of 0.78 mev. The tissue penetration expressed as half-value layer is clearly more than is required for the treatment of ophthalmic tumors. The short half-life (3.8 days) is an added inconvenience because it is likely that new radon seeds will be required for each patient. The high energy and short half-life necessitate delivery of the total dose at a very high dose rate. This results in excessive irradiation of the tissues near the radon. Finally, since the radon is supplied as a gas enclosed in metal seeds, the radiation safety presents certain drawbacks that have led many hospitals to discontinue its use.<sup>8,9</sup>

The effectiveness of cobalt-60 in the treatment of choroidal melanoma has been shown by Stallard.<sup>10</sup> He has reported a large series of patients over many years, and his five- and ten-year cures are comparable to or better than those from enucleation (Tables 1, 2). However, his case documentation did not have the benefits of indirect ophthalmoscopy, ultrasonography, infrared photography, fluorescein angiography, or radioactive phosphorus up-

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